

Kyle Main

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Senior Cybersecurity Engineer & Data Scientist — Detection Engineering, Security Operations, Data Platforms

Cybersecurity engineer and data scientist with 12 years of experience spanning detection engineering, security operations, machine learning, and data engineering — from building detection platforms and multi-SIEM orchestration frameworks from scratch to supporting live SOC incident response for federal agencies. Deep, cross-employer expertise in Elasticsearch-based SIEM/analytics platforms, full-lifecycle detection content (signature, statistical, behavioral, and ML-based), and large-scale data engineering (220+ ingested log sources, GCP Dataproc/Dataflow, PySpark/SparkSQL). Active federal security clearances (Top Secret, DOE Q, Public Trust) from direct work inside DOE, DOE/NNSA, and Treasury security programs.

SECURITY CLEARANCES

Top Secret — current, sponsored by U.S. Treasury | DOE Q Clearance — held | Public Trust — held, sponsored by DOE

COMPLETE SKILLS INVENTORY

Elasticsearch (deep, cross-employer): ES queries (Query DSL) for detection/investigation; ES transforms for detection content (built the UEBA layer at DOE/NNSA on top of transforms); Logstash (50+ custom parsing/normalization filters); ES Beats (multiple variants deployed for log collection); native ES detection rules / Watcher-style alerting; ES API (querying, index/cluster management); Kibana dashboarding and visualization

Cribl: Hands-on creation and management of Cribl pipelines (observability/data-routing product)

Core Detection Engineering & SIEM: ELK Stack, Splunk, MITRE ATT&CK, signature/statistical/behavioral/ML-based detection content development, PySpark, GCP Dataproc/BigQuery/Dataflow, AWS/GCP/Azure (Azure scoped to Microsoft Sentinel/Defender API orchestration), Git

Multi-SIEM Detection-as-Code & Orchestration: Built rule/content orchestration across Microsoft Sentinel, Microsoft Defender, Google SecOps (Chronicle), Splunk, CrowdStrike, SentinelOne, Sumo Logic, Palo Alto XSIAM, and Devo via native APIs, plus prior ArcSight experience; reusable per-technology API adapters covering rules, alerts, tables, and schemas; created and governed API tokens/roles/permissions across these platforms; implemented multithreaded parallel rule deployment across customers within a GitLab CI/CD pipeline; tracks detection rule quality metrics (coverage, precision/false-positive rate) with staged/safe production rollout; wrote automated unit/integration tests for the orchestration pipeline itself

Cloud IAM: Implemented IAM policies and roles in both AWS and GCP

Distributed Systems / Streaming: Kafka familiarity; hands-on exposure to Flink jobs in a production environment

Serverless / Event-Driven Architecture: Built event-driven enrichment on GCP — serverless functions triggered by incoming events to enrich data before it's written/forwarded downstream

Leadership: Team-lead and sprint-lead experience

Threat Intelligence Integration: Integrates CTI (indicators, TTPs, actor/campaign context) directly into detection rule logic; enriches security alerts/cases with threat intel context to speed up triage; uses CTI for investigation and false-positive research; sources include Vedere Labs (Forescout's in-house threat research team)

GenAI / LLM Applications for Security: Prompt engineering for security use cases — analyzing security data, identifying false positives, generating new detection content/rules; using GenAI to interact with SIEM APIs for detection-content orchestration; built reusable GenAI-powered "skills" for detection engineers to automate repetitive tasks (e.g., converting detection rules between SIEM rule syntaxes)

Containers / Orchestration: Docker — managed containers and images, built reusable custom images, used containers for reproducible detection-testing environments; worked within a Kubernetes-orchestrated platform as a user (not cluster administration)

IaC / DevOps: GitLab CI/CD as the backbone for detection-as-code pipelines

Vulnerability Management / Endpoint Security: Tenable — hands-on user; ingested Tenable vulnerability scan logs into a SIEM/analytics platform and built analytics/detection content on top of that data; vulnerability exposure analysis; security control evaluation across many vendors; Microsoft Intune (end-user/customer level)

Data Science / ML: Python, SQL, advanced mathematics, pandas, scikit-learn, NumPy, SciPy, PyTorch, clustering/unsupervised ML, time-series anomaly detection, R

Data Engineering: Built data pipelines ingesting logs from 220+ sources into SIEM/Elasticsearch across multiple employers; Apache Beam / GCP Dataflow for historical/cold-storage data retrieval; Common Information Model / data dictionary design and standardization

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint

Oct 2023 – Present

Remote — Dallas/Ft. Worth Area. Three sequential federal security projects:

- Treasury SOC (TSSOC), Threat & Research team — CURRENT: Creates and manages analytics (Splunk saved searches) that serve as the SOC's detection and alerting content, analytically supporting Treasury's security incident/case work.
- DOE/NNSA Security Data Integration (SDI) — COMPLETED: Built an entirely new system ingesting CrowdStrike, Suricata, and Zeek telemetry into a central Elasticsearch platform, and built the analytics/detection layer on top of it — custom dashboards, data transforms, UEBA (User and Entity Behavior Analytics) detection content built on the transform outputs, and data-quality monitoring content and alerting.
- CISA CDM (Continuous Diagnostics and Mitigation) at DOE — COMPLETED: Supported data ingestion and data-quality efforts within an Elasticsearch and Splunk environment for a federal continuous-monitoring program.

Senior Threat Detection Engineer and Data Scientist — Forescout

Aug 2022 – Oct 2023

Remote — Dallas/Ft. Worth Area. Cysiv was acquired by Forescout in August 2022; continuation of the Cysiv role below.

- Senior member of the data science and threat detection engineering team creating content to detect cyber threats in massive customer data — cloud-based big data tools, signature/behavioral/statistical/time-series/ML-based detection rules.

Threat Detection Engineer and Data Scientist — Trend Micro / Cysiv

Sep 2018 – Aug 2022

On-site — Irving, TX. Cysiv started as an internal Trend Micro team project, spun out as its own company in 2020. Very early hire.

- Built out the rules engine, detection content, and data engineering for the startup from scratch — data engineering/pipelining for 220+ unique log data sources, detection content in a next-gen cloud-based SIEM, 2,300+ individual detection rules covering most of the MITRE ATT&CK matrix, and 50+ data filters.
- Detection rule development across signature-based, statistical-based, behavioral, aggregation/threshold, and ML-based rules (clustering devices on the network by feature).
- Time-series anomaly detection of entity behaviors: Outlook process chains; child/parent process combinations and chains; authentication behaviors (auth attempts by country over time, anomalous volume/attempt detection).
- Exploratory data analysis at scale on GCP Dataproc compute clusters via Zeppelin notebooks, using a home-grown reusable analysis toolkit and Spark jobs to load data from buckets, with PySpark/SparkSQL for analysis.
- Data engineering: 50+ Logstash filters for parsing/normalization; deployed Elasticsearch Beats for log collection; built a Common Information Model (CIM) — a data dictionary standardizing field names/types across all parsed data; assisted building "Loggify," a homegrown log parsing/filtering tool that replaced Logstash; connector/collector health monitoring and troubleshooting; fetched large volumes of historical cold-storage data for customers using a homegrown Apache Beam program run via GCP Dataflow.

Security Data Scientist — Experian

Jan 2015 – Jan 2018

On-site — Allen, TX.

- Provided analytical support and guidance to Information Security teams, analyzing large datasets to develop custom models and algorithms supporting identification and handling of emerging threats.
- DNS-based detection and mitigation of malware infections on a network.
- Security log data analysis for discovery of anomalous behavior.

EDUCATION

M.S. Physics — University of North Texas, Dec 2013 (Optics/Photonics; Numerical Data Analysis & Modeling; Organic/Inorganic Materials — FRET mechanism; Applied Physics). Graduate Assistant Tuition Scholarship, Competitive Tuition Waiver.

B.S. Physics — Ball State University, May 2011. Minors: Mathematics, Astrophysics. Thesis: "Luminescence Properties and a Whispering Gallery Mode Microlaser in AlN:Ti." SPS Physics Honor Society, Dean's List.

CERTIFICATIONS

Splunk User Certification — Splunk Education, June 2017 | Splunk for Analytics and Data Science — Splunk Education, Jan 2017

R Programming — Johns Hopkins via Coursera, Aug 2015 | The Data Scientist's Toolbox — Johns Hopkins via Coursera, Aug 2015